

$$Y = ZK^{.3}N^{.7} \quad \left. \vphantom{Y = ZK^{.3}N^{.7}} \right\} \text{Cobb-Douglas with CRS}$$

$$\ln Y = \ln[ZK^{.3}N^{.7}]$$

$$= \ln Z + \ln K^{.3} + \ln N^{.7}$$

$$\ln Y = \ln Z + .3 \ln K + .7 \ln N$$

how does $\ln Y$ change as Δ 's "change"

$$\left(\frac{d \ln Y}{d Y} \right) \Delta Y$$

$$\left(\frac{1}{Y} \right) \Delta Y = \frac{\Delta Y}{Y} = \% \Delta \ln Y$$

$$\% \Delta Y = \% \Delta Z + .3 \% \Delta K + .7 \% \Delta N$$

growth accounting

$\nabla \Delta Y$ } comes from
(requires) a $\nabla \Delta$
in Z, K , and/or N

$\Rightarrow Y, K, Z, N$ } more
together

"trend" } over long
term and over

"business
cycle" } short run



Macroeconomic Data: Trends, Cycles, and the stylized facts of the Business Cycle

Data are hyperlinked to original sources

Reference: Chapters 1-3, Williamson

- Skim Chapters 1-2

-  Things you can skip from Chapter 3:
- The sections on “The Price Level and Inflation” and “Seasonal Adjustment”

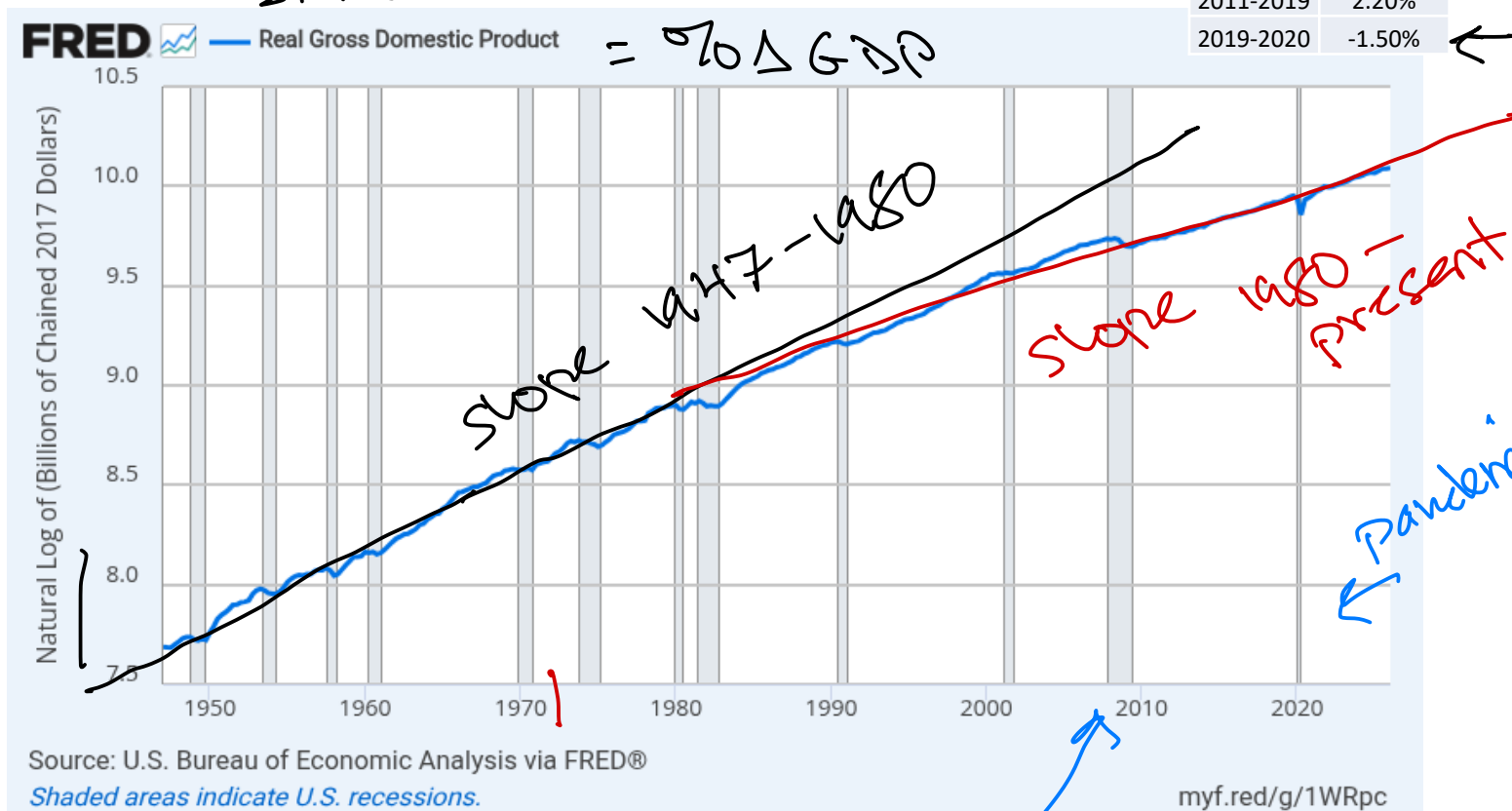
Trend growth was higher in the post-war 20th Century than it has been recently, even excluding the Pandemic

$$\text{Slope} = \frac{\Delta \ln \text{GDP}}{\Delta \text{time}} = 1 = \frac{\Delta \ln \text{GDP}}{\Delta \text{GDP}} \Delta \text{GDP} = \% \Delta \text{GDP}$$

1951-1970	3.80%
1971-1990	3.30%
1991-2000	3.60%
2001-2010	1.80%
2011-2020	1.14%
2021-2025	2.50%
2011-2019	2.20%
2019-2020	-1.50%



Slope = growth rate



Global Financial Crisis

Recessions were more frequent in the post-war 20th Century than they have been recently

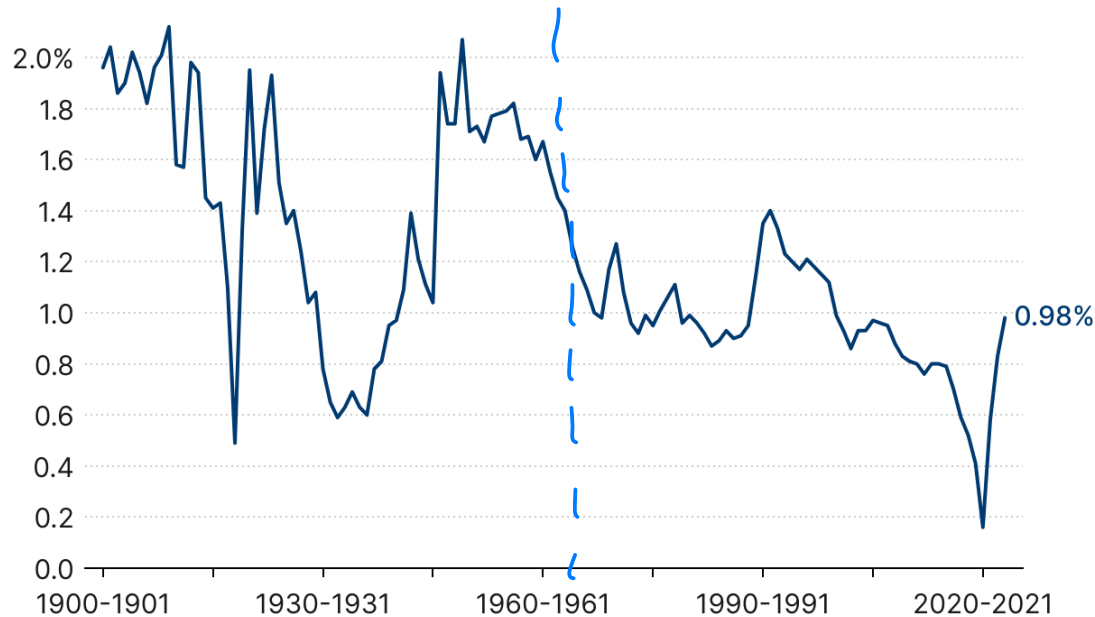


Trend GDP is flatter in part because trend population growth has been flatter

Baby Boom 1950-1964

US annual population growth

Years 1900-01 to 2023-24*



Source: William H. Frey analysis of Census Bureau historical population estimates including 2020-24 annual estimates, released December 19, 2024

Note: *July 1 to June 30 of each year

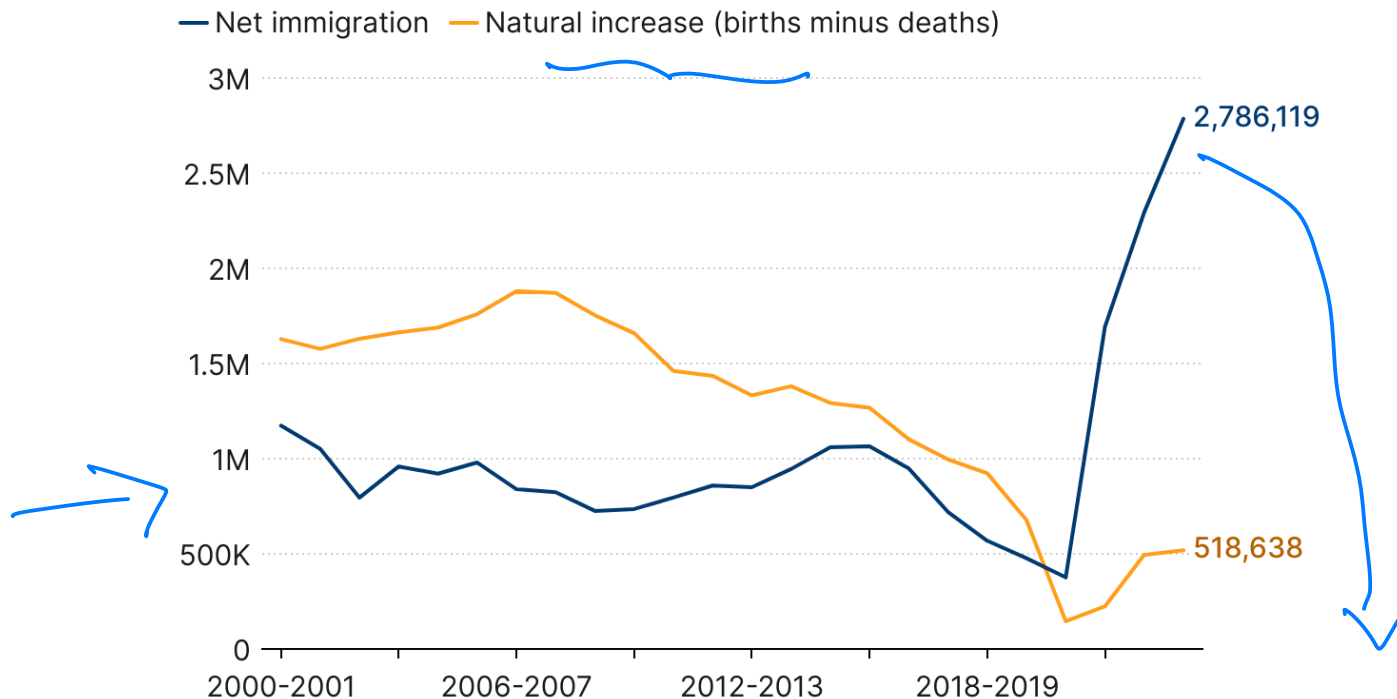
B | Brookings Metro



Without immigration, future trend population and GDP growth will be even flatter

US net immigration and natural increase

2000-01 through 2023-24



Source: William H. Frey analysis of Census Bureau population estimates: vintage 2020 and vintage 2024, released December 19, 2024

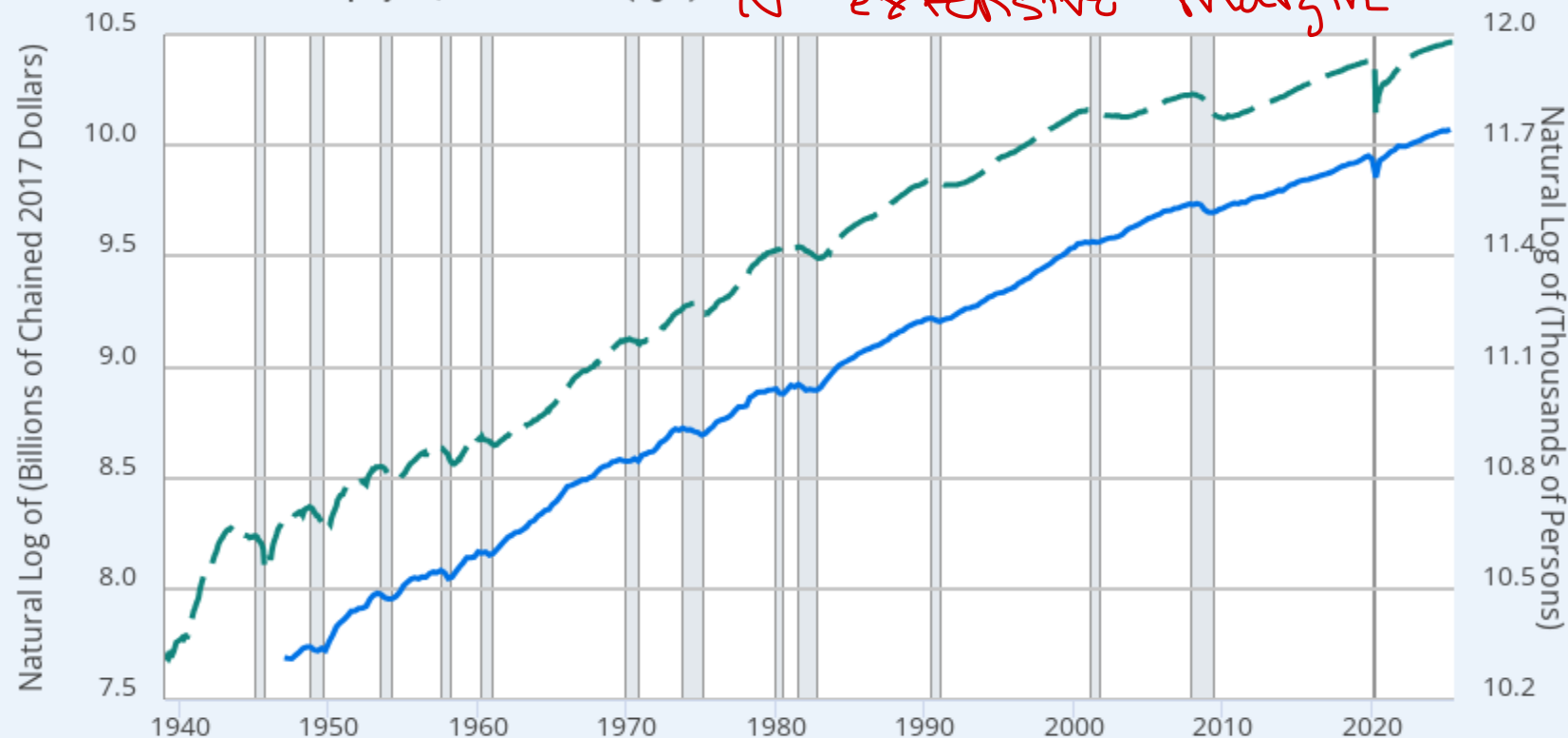
Note: *July 1 to June 30 of each year



FRED

Real Gross Domestic Product (left)
All Employees, Total Nonfarm (right)

Y
N extensive margin



Sources: U.S. Bureau of Economic Analysis; U.S. Bureau of Labor Statistics via FRED®

Shaded areas indicate U.S. recessions.

myf.red/g/1LMYr

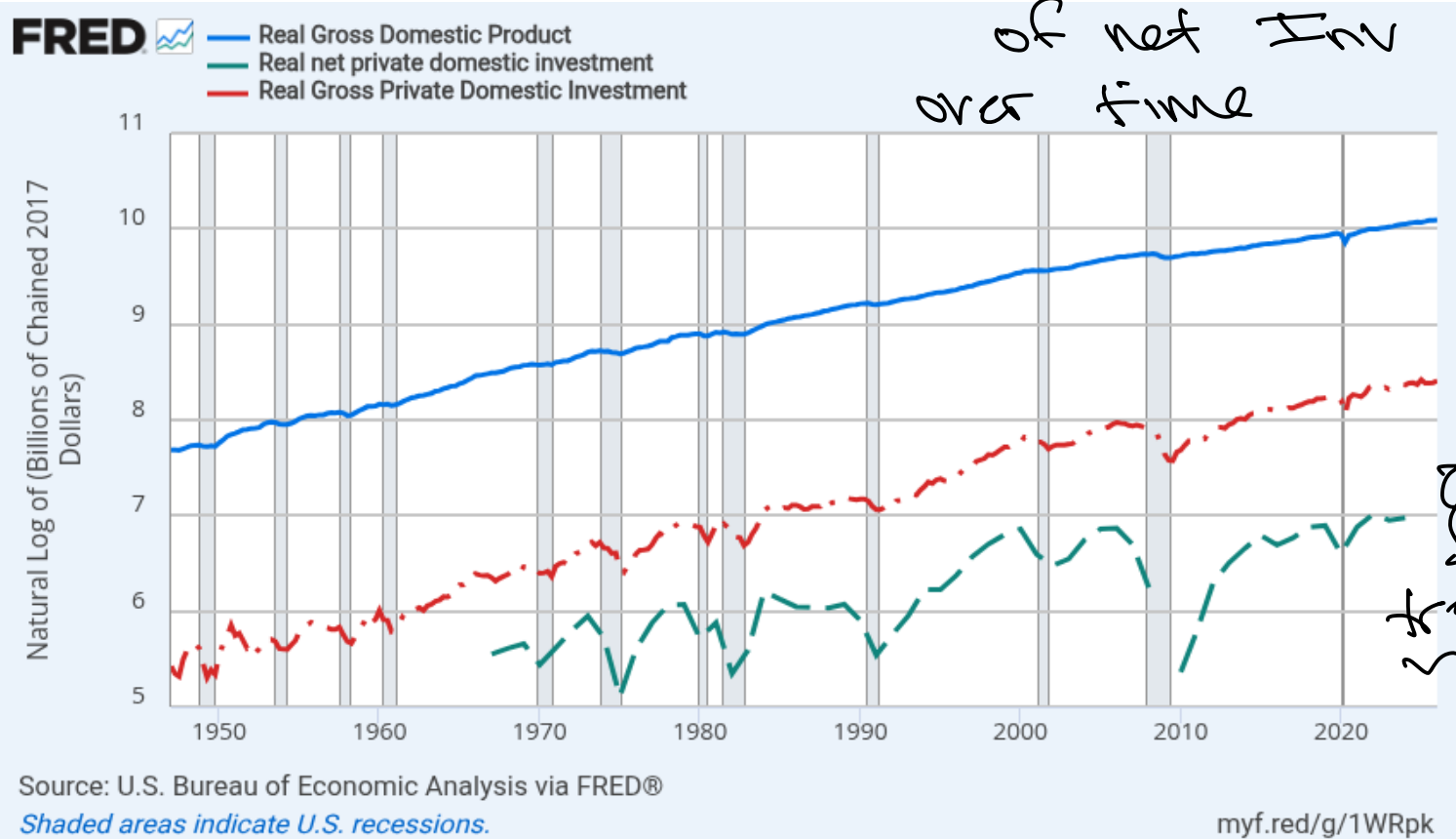
trends and cycles correlated
ch 4, 5, 6 } N

"Secular stagnation"

Net Investment has also been fairly flat,
with large cyclical fluctuations

K's sum

of net Inv
over time



no
trend
12 flat

Trend Inv and GDP spending
correlated (more together) as is
the cycle.

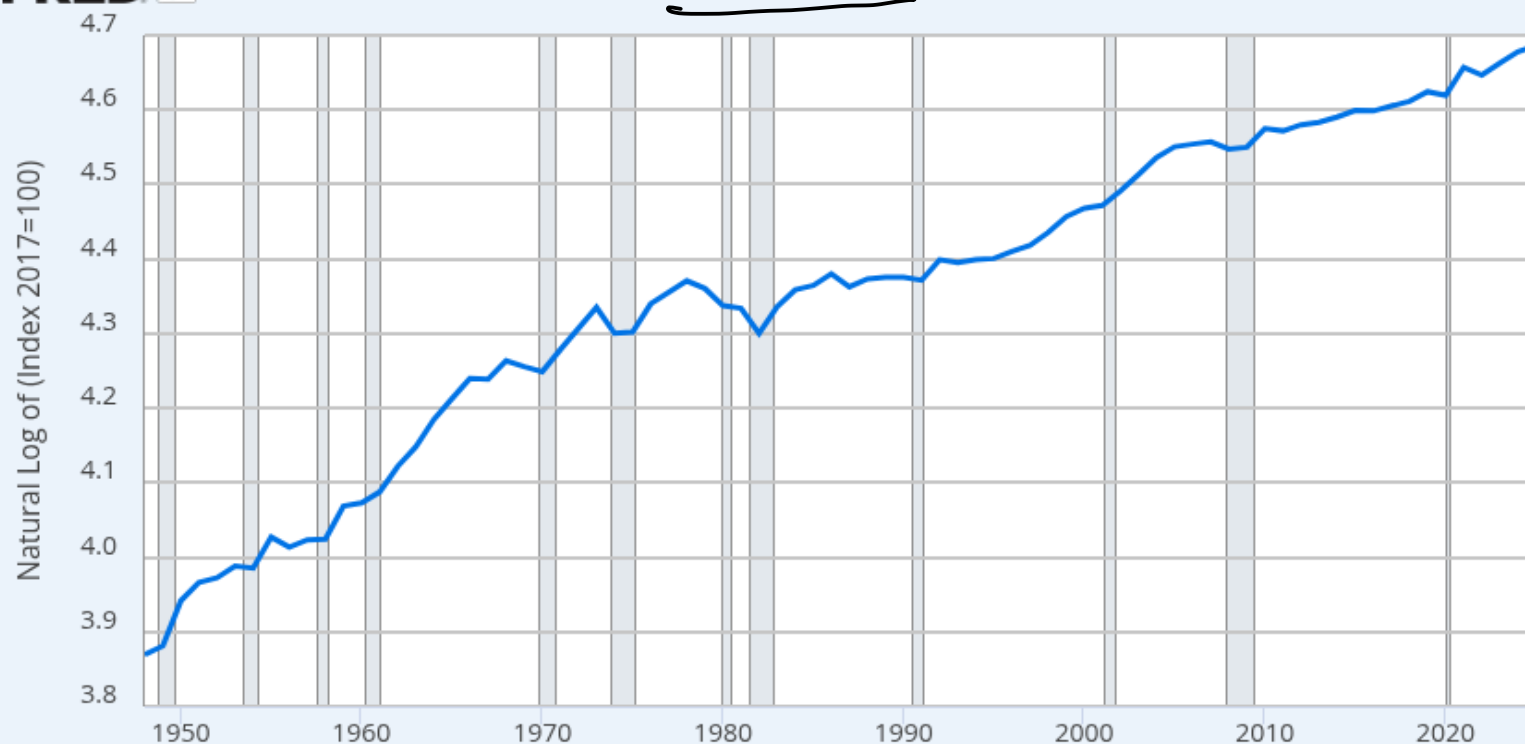
Combined capital and labor productivity growth has also slowed and also moves cyclically with GDP



"Z" or "TFP"

FRED

Private Nonfarm Business Sector: Total Factor Productivity



Source: U.S. Bureau of Labor Statistics via FRED®

Shaded areas indicate U.S. recessions.

myf.red/g/1WRpl

ch 4-7 } look at TFP



Business cycles



Figure 3.1

Idealized Business Cycles

Peak: High point
before GDP falls
back to trend.

Trough: low point
before GDP rises
to trend

Amplitude: how
far do we get
from trend

Peak to Trough:
Recession

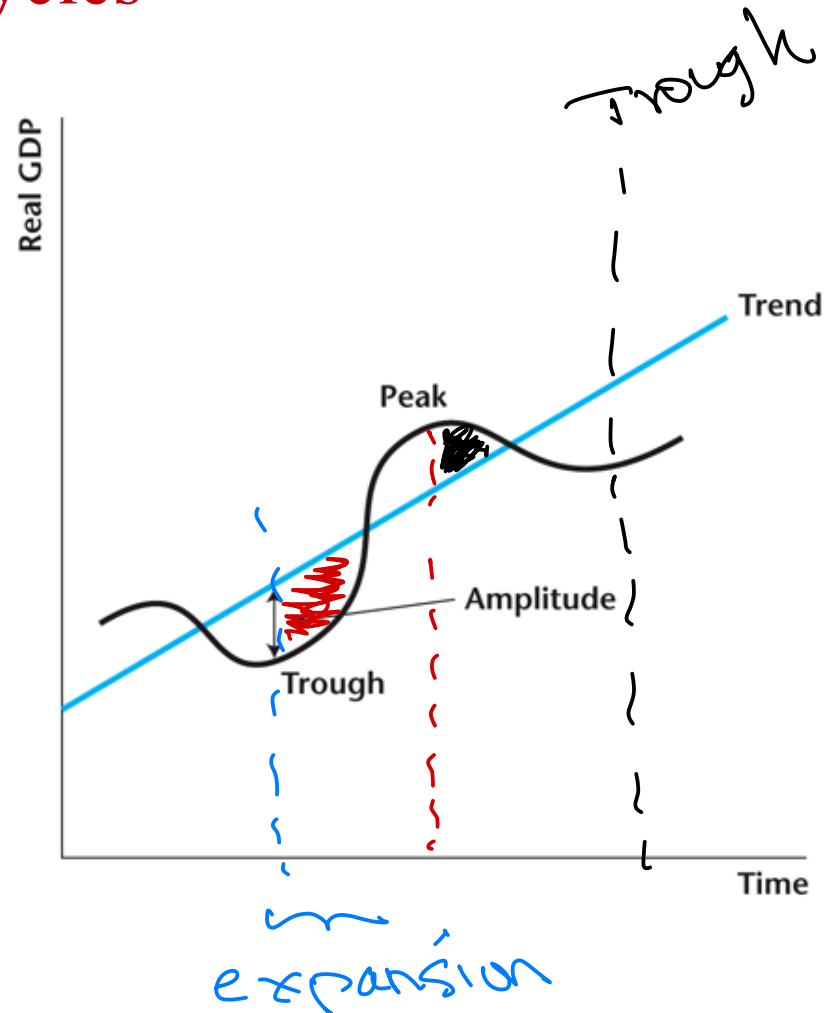
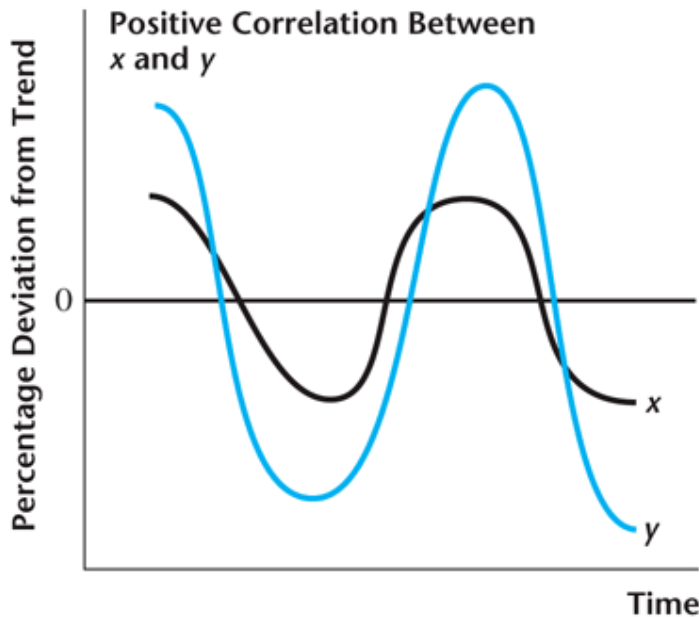




Figure 3.3

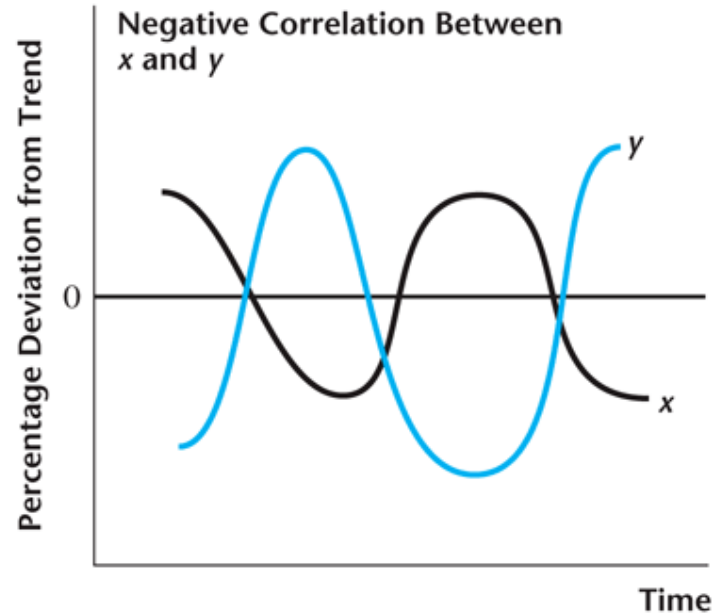
Time Series Plots of x and y

unemployment $\uparrow$
as GDP $\downarrow$



(a)

move together
(max = 100%)



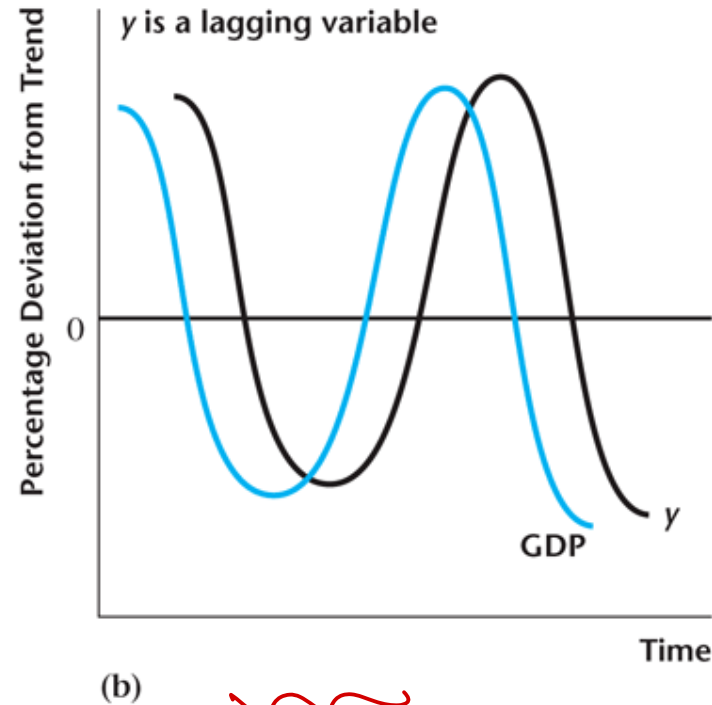
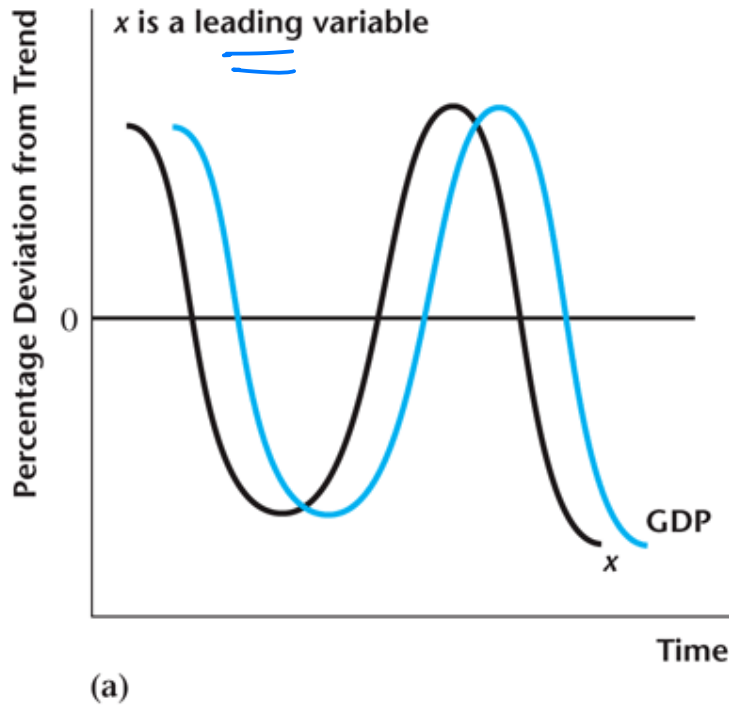
(b)



Figure 3.7

Leading and Lagging Variables

} relative
to
GDP



employment lags
GDP (ch 6)



Table 3.1

Correlation Coefficients and Variability of Percentage Deviations from Trend

	Correlation Coefficient	Standard Deviation (% of S.D. of GDP)	
Consumption	0.77		77
Investment	0.80	<i>amplitude</i>	* 301
Employment	0.78		65
Average Labor Productivity	<i>z</i> 0.77		63

~~*~~ Golden table (ch 5)

Table 3.2 Summary of Business Cycle Facts

"stylized" Facts

	Cyclicalities	Lead/Lag	Variation Relative to GDP
Consumption	Procyclical	Coincident	Smaller
Investment	Procyclical	Coincident	Larger
Employment	Procyclical	Lagging	Smaller
Real Wage	Procyclical	?	?
Average Labor Productivity	Procyclical	Coincident	Smaller



Some issues with our data:

**Estimations of labor share in the Aggregate
Production Function seems to be settling at a
lower level**

Said: $\gamma = z K^{-3} N^{-7}$

But since 2000 ...

$$\gamma = z K^{-4} N^{-6}$$

(automation)

Summarizes next slide

Marginal vs Average Productivity



Logically, $MPN = w$, but there are data issues:

- We only have *average* productivity and compensation data.
- During recessions, low wage workers are more likely to become unemployed, causing the average earned wage to rise because of “compositional effects” rather than wage increases *per se*

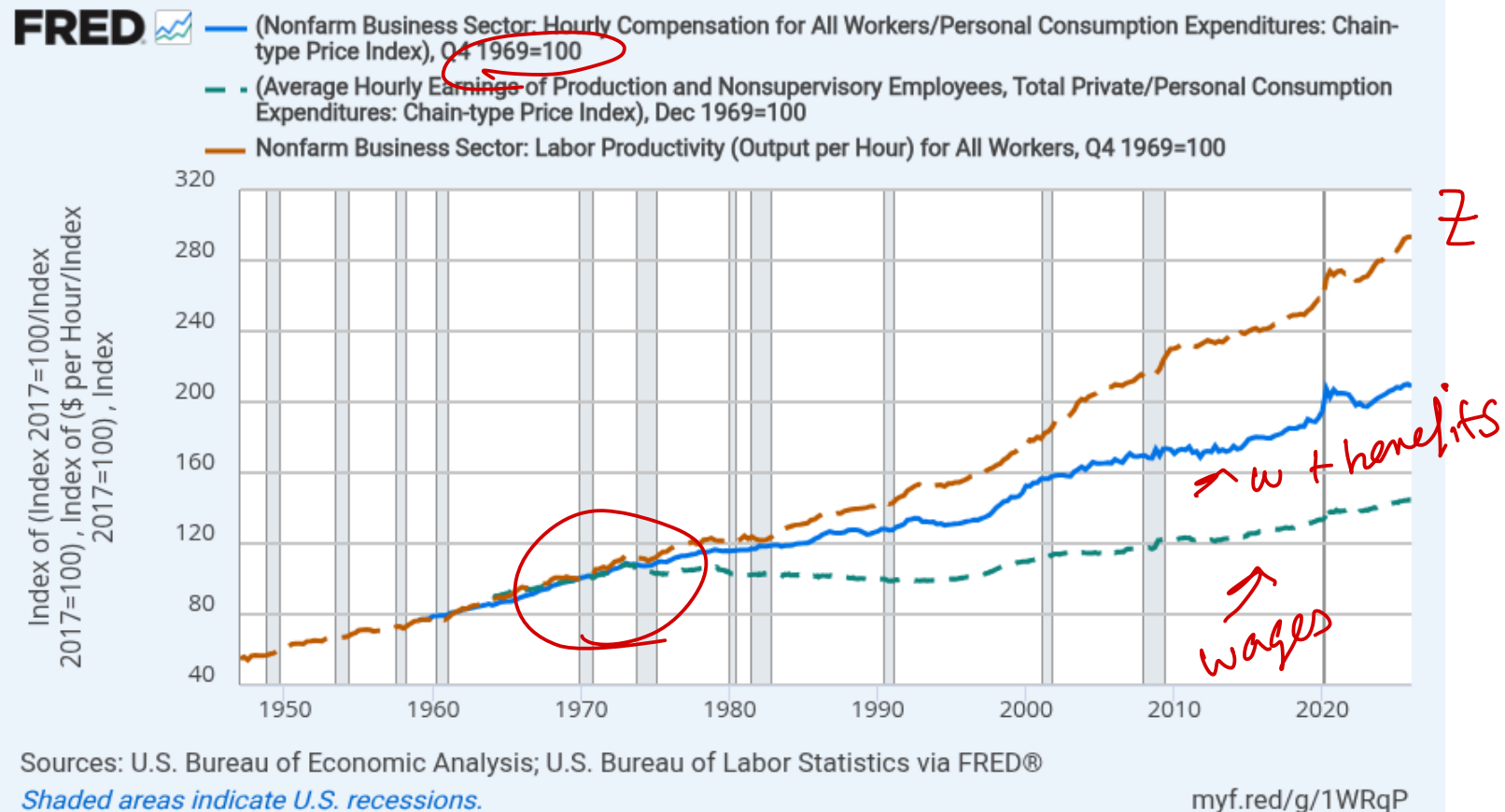
Over the longer term, we observe:

Since 1970,

- Worker productivity has been rising faster than compensation
- Most of the increase in compensation has been in the form of benefits rather than in take-home pay, with take-home pay rising much more slowly than total compensation
- Most of the disparity between take-home pay and total compensation happened prior to the Great Recession, possibly reflecting longer-term (trend) compositional effects.

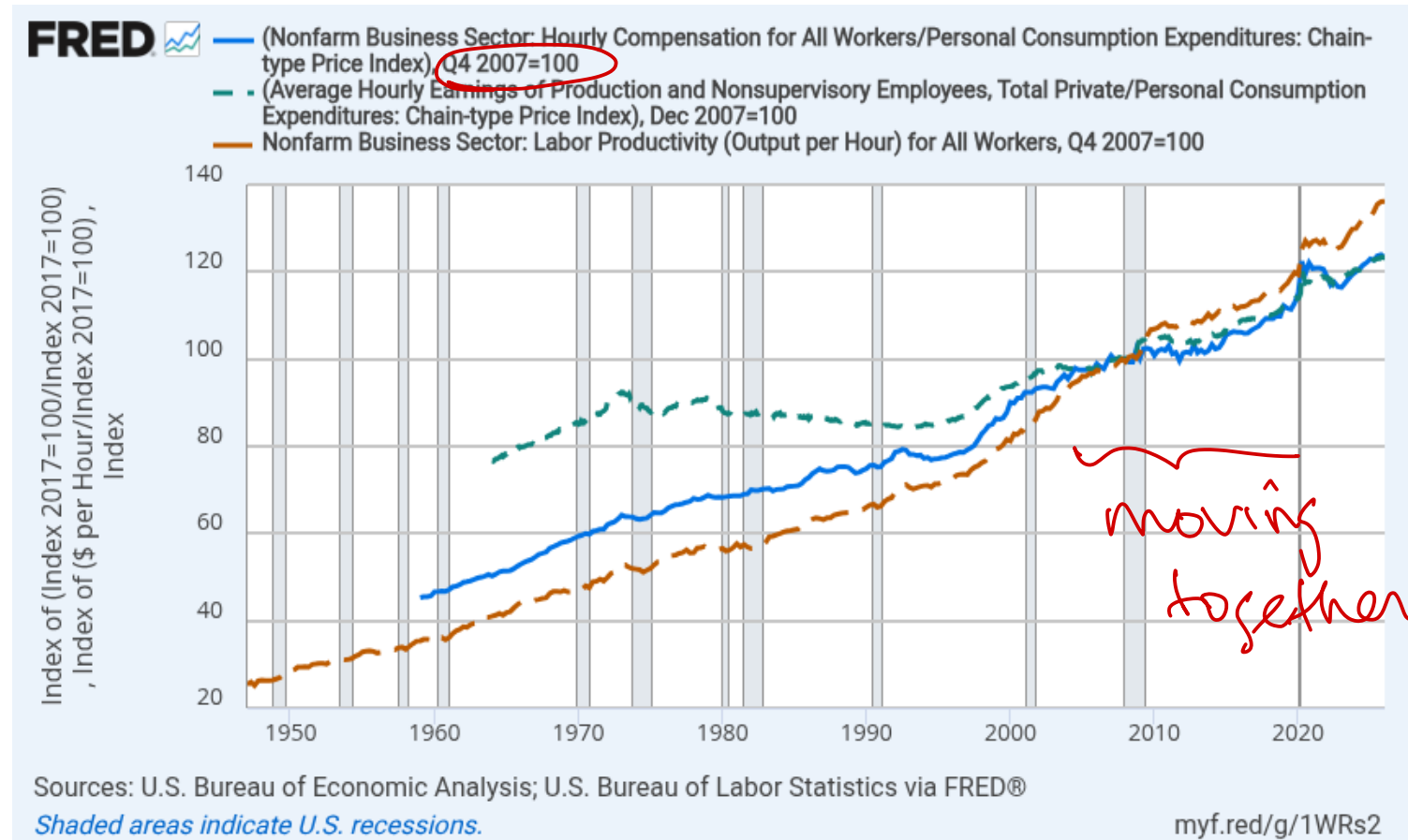


1970-2026: **Worker productivity** almost tripled, while **total compensation** more than doubled. **Take-home pay** was fairly flat until the mid-90's and has been rising steadily, though slowly, since, for a gain of 44% since 1970.





Compensation and earnings have been moving together since the Great Recession.



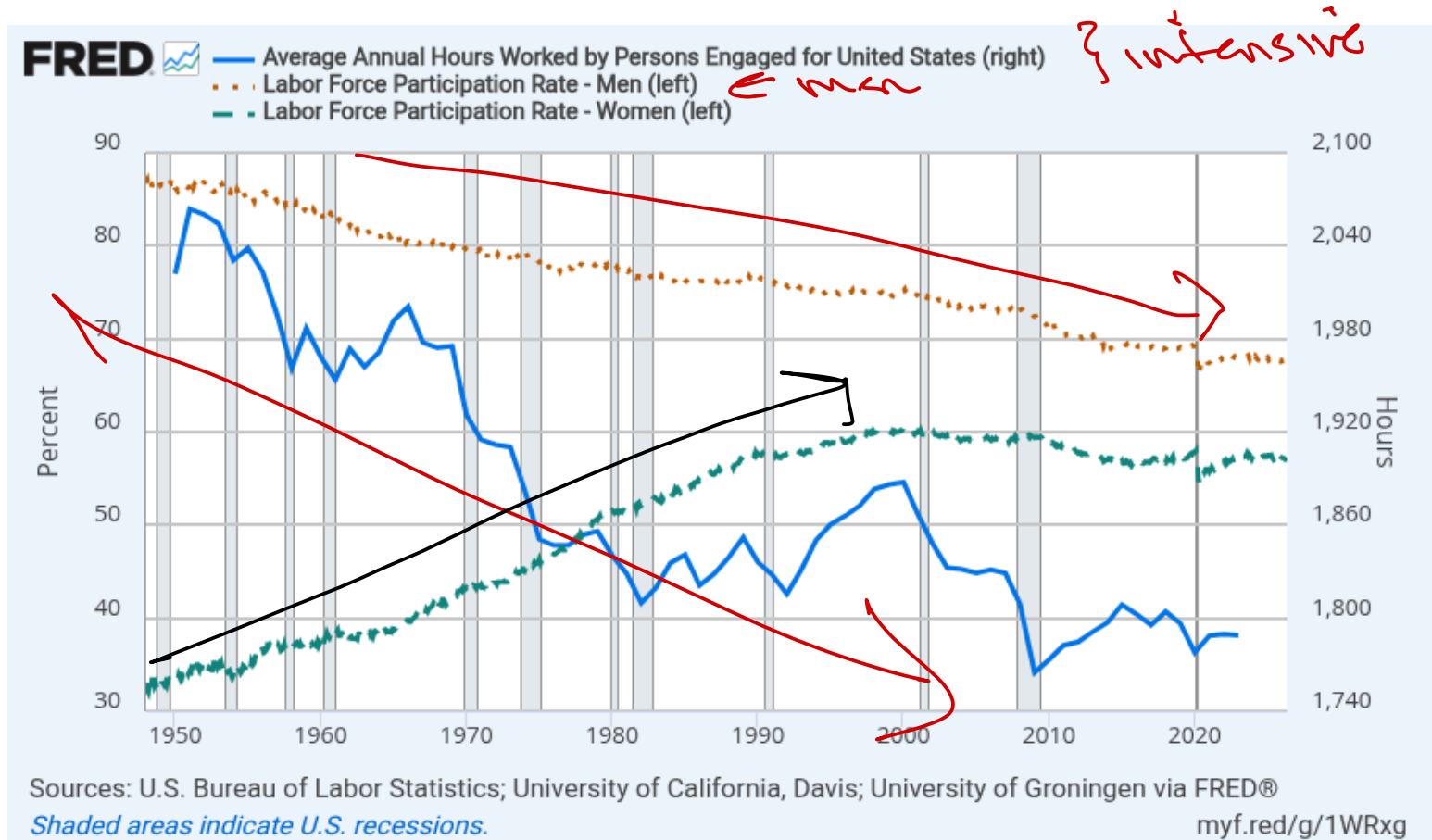


Some issues with our data: Ambiguity about the slope of labor supply


* ch 4



At the intensive margin, hours worked are falling as real compensation is rising, but there is no such pattern at the extensive (participation) margin



NS downward sloping at intensive margin?!



At the intensive margin, hours worked are falling as real compensation is rising, but there is no such pattern at the extensive (participation) margin

